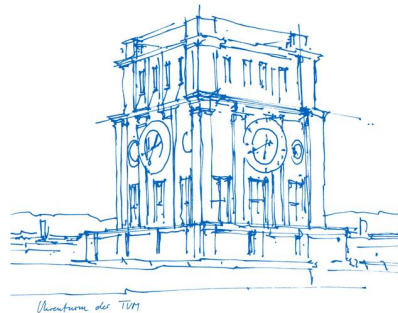


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in
DEM-generated zeolite beds

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Introduce the project: how random packing geometry can inform a reduced model of transient adsorption.



Presentation outline

Motivation and research question

Pore-network reference model

Reduction and diffusivity prediction

Results and robustness

Conclusions and next steps



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└ Presentation outline

We first motivate inlet accessibility, then explain the PNM reference and its mean uptake output. Next we reduce it to a homogeneous bed and develop the diffusivity closure. Results and conclusions follow.

Presentation outline

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Pore-network reference model

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Conclusions and next steps

Section 1 of 5

Motivation and research question

Why should inlet-connected pathways matter?

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Why examine access from the inlet?

What porosity misses

- Porosity measures how much void space exists.
- It does not describe bottlenecks, path lengths or parallel connections.
- Similar void fractions can therefore hide different gas-supply pathways.

Application: CO₂ adsorption in zeolite 13X, used here as a controlled model problem.

Hypothesis: inlet-connected diffusive pathways contain information that porosity misses.

Why the inlet is the reference

- Gas enters from a reservoir below the bed.
- Adsorption removes gas as it diffuses into the packing.
- Access from the inlet may therefore help explain transient uptake differences.

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└ Motivation and research question

└ Why examine access from the inlet?

The motivation is transport through random geometry. Two packings can have similar porosity but very different arrangements of constrictions and parallel pathways. In our experiment gas is supplied at the bottom and removed by adsorption as it enters. That makes the inlet a physically meaningful reference boundary. This leads us to test inlet-to-interior accessibility as a descriptor. It is a hypothesis, not a claim that the inlet alone controls every time scale. CO₂ and zeolite provide a controlled application for studying this question.

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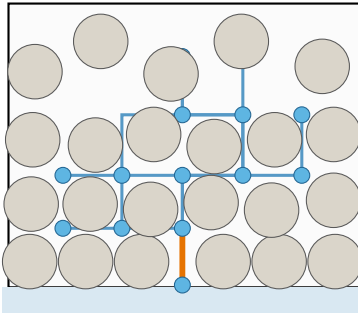
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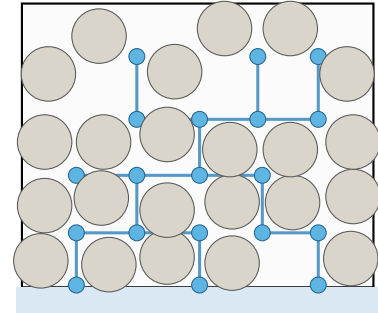
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



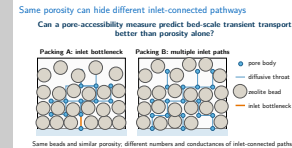
- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

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└ Motivation and research question

└ Same porosity can hide different inlet-connected pathways



This schematic introduces the central problem. The two beds contain the same monodisperse beads and can have almost the same total porosity. Their inlet-connected paths can nevertheless differ strongly. Packing A illustrates a network reached through one restrictive inlet bottleneck. Packing B illustrates several paths that connect the reservoir to the bed interior.

The blue lines show only simplified inlet-connected paths, not every throat in a real extracted network. The sketches are therefore hypotheses rather than simulation results. They motivate our research question: can a quantitative pore-accessibility measure predict transient bed-scale transport better than bulk porosity alone?

Section 2 of 5

Pore-network reference model

From particle geometry to the mean uptake curve

From particle geometry to the mean uptake curve.

Twenty beds isolate variability caused by random arrangement



Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m ⁻³ *
simulation time	5000 s

Controlled diffusion–adsorption problem

- same zeolite beads and tube, only the insertion seed changes
- dry, isothermal gas supplied from a reservoir below the bed
- interparticle diffusion and bead adsorption, with no imposed flow
- heat effects and competitive water adsorption are outside this model

* Approximately 15 vol.% CO₂ at atmospheric pressure and 298 K.

The ensemble tests arrangement effects under otherwise identical conditions.

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The ensemble tests arrangement effects under otherwise identical conditions.

We isolate arrangement effects using twenty packings of identical zeolite beads in the same tube. Temperature, inlet concentration and adsorption parameters are fixed. The gas is dry and isothermal, with no imposed through-flow. The concentration is approximately equivalent to fifteen volume percent CO₂ at atmospheric pressure and 298 K. This is a controlled diffusion-driven adsorption problem, not an industrial breakthrough simulation. The next step is to turn the particle geometry into a pore network.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its valid circumcentre represents a pore node
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels
- 2^{22} Sobol points sample the solid and gas space.
- Gas-point counts allocate the calculated void volume to pores.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

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- └ Pore-network reference model
- └ Pore-network geometry and throat diffusion

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The particle centres define Delaunay tetrahedra. Valid circumcentres become pore nodes, and a shared triangular face defines a candidate throat between neighbouring nodes. The throat conductance follows the molecular diffusivity times an approximate opening area divided by length. Sobol sampling assigns gas volumes to nodes, while connectivity comes from geometry. This is the construction summary; the detailed sketches remain in the optional appendix. Now we have storage volumes and connections and can write a gas balance for every pore.

The pore network solves diffusion and adsorption together

Gas balance for each pore i

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - S_i.$$

- V_i : gas volume, c_i : gas concentration
- G_{ij} : throat conductance [$\text{m}^3 \text{s}^{-1}$]
- G_{iR} : connection to reservoir at c_R , zero away from the inlet
- S_i : net gas transfer to adjacent beads [mol s^{-1}]

Uptake for each bead k

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

- q_k : current loading [mol kg^{-1}]
- k_{LDF} [s^{-1}]: linear-driving-force rate coefficient
- $1/k_{\text{LDF}}$: relaxation time at fixed gas concentration

One coupled system: one concentration per pore and one loading per bead.

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This is the transient diffusion equation on the network, expressed as a balance on each pore volume. The first term exchanges gas with neighbouring pores, the second connects inlet pores to the finite-conductance reservoir, and the final term transfers gas to beads. Each bead has one loading equation even if several pores surround it. LDF means linear driving force: the rate is proportional to the difference between equilibrium and current loading. All pore concentrations and bead loadings are integrated together from the prescribed initial state. The total transfer satisfies $\sum_i S_i = \sum_k m_k dq_k/dt$, so adsorption transfers gas into the solid inventory. In the code, positive uptake is allocated among incident pores using incidence weights multiplied by positive gas concentration, preventing empty pores from supplying gas. Desorption uses normalized incidence weights. No independent gas-flow or pressure equation is solved.

Local equilibrium sets the adsorption target

Local equilibrium loading

$$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

- c_k : average gas concentration around bead k
- p_k : local CO₂ partial pressure [Pa]
- q_k^* : equilibrium target [mol kg⁻¹]

Fixed material parameters

- q_s : maximum CO₂ loading [mol kg⁻¹]
- b : Langmuir affinity [Pa⁻¹], so $b p_k$ is dimensionless
- k_{LDF} : kinetic coefficient [s⁻¹]

Actual loading evolves in time

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

Local gas concentrations set the target. Kinetics determine how quickly beads approach it.

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The local gas concentration around bead k is a normalized, pore-volume-weighted average over its associated pores. The implementation uses nonnegative concentrations in this calculation. The ideal-gas relation gives the local CO₂ partial pressure and the Langmuir isotherm supplies the equilibrium loading. This equilibrium target is not a new differential unknown. The actual loading evolves through the LDF equation introduced on the preceding slide. The saturation loading and affinity are prescribed material parameters, and k_{LDF} is fixed across packings. Adsorption removes gas from the incident pores and adds it to the bead, preserving the combined inventory apart from reservoir exchange.

The pore network delivers the reference uptake curve

Resolved transient outputs

- Gas concentration $c_i(t)$ in every pore
- Adsorbed loading $q_k(t)$ of every bead
- Different bead loadings reveal incomplete utilization

The detailed model retains the actual pore connections and bead associations.

Bed-average uptake

$$\bar{q}_{\text{PNM}}(t) = \frac{\sum_k m_k q_k(t)}{\sum_k m_k}.$$

For identical beads:

$$\bar{q}_{\text{PNM}}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t).$$

This curve is the reference for fitting the reduced 1D model.

Next: can a homogeneous 1D bed reproduce this complete uptake curve?

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The network calculation produces many concentration and loading histories. For model reduction, we summarize the adsorbed inventory as the mass-weighted mean loading of all beads. Because the beads have equal masses, this is their arithmetic mean. We retain the full time series, not only the final loading. The PNM curve is a numerical reference under our modelling assumptions, not an experimental measurement. We now ask whether a much smaller homogeneous model can reproduce this curve.

Section 3 of 5

Reduction and diffusivity prediction

Fit a reduced model, then predict its coefficient

Fit a reduced model, then predict its coefficient.

Reduce the detailed network to a homogeneous 1D bed

Detailed pore-network model

- Actual pores and throats from each packing
- One gas concentration per pore
- One loading per bead
- Reference mean uptake $\bar{q}_{\text{PNM}}(t)$

Homogeneous 1D model

- Replace the network by 100 equal axial slices
- Each slice spans the full tube cross-section
- One gas concentration and one mean solid loading per slice
- Represent unresolved pathways with a single diffusivity D

Fit D so the reduced model reproduces the network's mean uptake history.

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The one-dimensional cells do not come from the pore-network extraction. We create a separate grid by dividing bed height into one hundred equal slices spanning the tube cross-section. Each slice contains homogenized gas and adsorbent phases. The detailed pore paths disappear from this representation. Their impact on transient uptake is represented by one fitted diffusivity. The reduced system has two hundred concentration and loading states, while the PNM has a state for every pore and bead. We compare their mean uptake curves rather than attempting a one-to-one mapping of their internal fields.

Reduction and diffusivity prediction

The homogeneous 1D uptake model

This slide describes the homogeneous one-dimensional model, not the individual pore network. Each axial cell contains gas and a homogenized amount of adsorbent. We solve one gas concentration and one solid loading per cell, giving 200 coupled state equations for 100 cells. The loading equation explicitly substitutes the Langmuir equilibrium relation and p equals cRT . The diffusivity appears in the gas balance. Changing it changes the gas concentration history and therefore the adsorption history. The right-hand side is a spatial average of the calculated loadings, not an additional independent evolution equation. For equal cells and uniform adsorbent density, the arithmetic average is the bed mean loading.

The homogeneous 1D uptake model	
Gas balance in each axial cell	Average the calculated loading
$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}$	$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz$
Adsorbent loading in each axial cell	For 100 equal axial cells:
$\frac{\partial q}{\partial t} = k_{LDF} \left[\frac{q_s b R T c}{1 + b R T c} - q \right]$	$\bar{q}_{1D}(t; D) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D)$
- Divide the bed height into 100 equal slices, each spanning the tube cross-section. - Solve gas concentration and mean solid loading together in each slice. - Trial D changes c, which changes adsorption.	- Match bed height, porosity and inlet condition to the network. - Keep isotherm, kinetics and initial state fixed. Only D changes.
The 1D grid is separate from the pore network.	

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Fit the effective diffusivity to the complete uptake curve

Per-bed optimization

1. Select a trial D within prescribed bounds.
2. Solve the coupled 1D equations from the same initial state.
3. Compare the full uptake curve with the PNM reference.
4. Use previous errors to select the next D and repeat (optimization).

Curve error and fitted coefficient

$$e_m(D) = \bar{q}_{1D}(t_m; D) - \bar{q}_{PNM}(t_m).$$

$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{PNM}(t_f)},$$

$$\sum_m w_m = 1.$$

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The time weights account for unequal spacing of saved outputs.

A fitted D_{eff} is a reduction target for each bed, not yet a geometry-based prediction.

Reduction and diffusivity prediction

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3. Compare the full uptake curve with the PNM reference.	$\sum_m w_m = 1.$
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The time weights account for unequal spacing of saved outputs.

A fitted D_{eff} is a reduction target for each bed, not yet a geometry-based prediction.

For each bed, we repeatedly solve the homogeneous model with different diffusivities. NRMSE means normalized root-mean-square error. Here we weight squared errors by elapsed time using trapezoidal weights, take the square root and divide by the final reference loading. This prevents densely saved times from dominating the fit. The bounded scalar optimizer operates in $\log_{10} D$; it starts deterministically and previous errors guide later trials. Only D changes. This produces one reduction target per packing. The next question is whether a cheaper calculation from geometry can predict that target.

Can inlet accessibility predict the fitted diffusivity?

A steady geometric descriptor

- Use the same pore network with adsorption disabled.
- Fix concentrations at the inlet and an interior plane.
- Solve steady diffusion and sum the resulting molar flux $\dot{n}$.

$$G_{\text{access}} = \frac{\dot{n}}{\Delta c}, \quad D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

H : bed height; A_{tube} : tube cross-section.

Accessibility measures steady gas access. The closure links it to transient uptake.

Learn a transport closure

- Pair each D_{access} with its fitted D_{eff} .
- Fit $D_{\text{eff}} = f(D_{\text{access}})$ on training beds.
- Leave one bed out to test diffusivity prediction.
- For a new bed, apply the closure and run the 1D model.

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└ Reduction and diffusivity prediction

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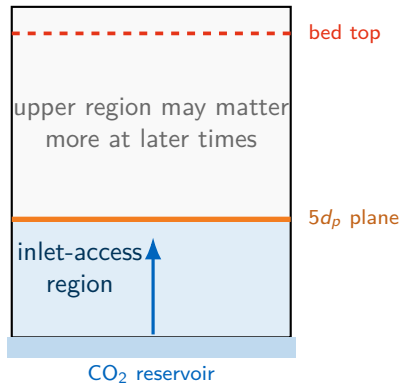
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Learn a transport closure

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- Leave one bed out to test diffusivity prediction.
- For a new bed, apply the closure and run the 1D model.

We return to the inlet-accessibility hypothesis. On each network we perform a separate steady diffusion calculation without adsorption, imposing a concentration difference from the inlet to a selected interior plane. Total molar flux divided by that difference defines the conductance. We multiply by full bed height divided by tube cross-section to form the diffusivity-scaled descriptor used in the code. This convention is distinct from measuring a local diffusivity over only the plane distance. It is not automatically equal to the fitted effective diffusivity: one describes steady access, the other reduces a transient adsorption problem. Their relationship is learned across beds. Leave-one-bed-out fitting uses nineteen beds to predict the remaining bed. This is internal validation in the current ensemble, not proof of universal transferability.

Why use an interior plane?



A plane at 5d_p includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks that may matter more at later times

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└ Reduction and diffusivity prediction

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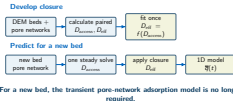
- measure conductance through the entire bed
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The plane provides a tunable spatial scale for the accessibility diagnostic. A near-inlet plane includes the entrance openings and some deeper constrictions and parallel paths. Using the bed top instead measures conductance through the complete network. The focus on a near-inlet region is motivated by entry-driven uptake, but it does not establish that upper pores are irrelevant at early times. In the current implementation, depth is measured above the lowest retained pore. Five bead diameters is the selected diagnostic depth. The results also compare three, seven and ten bead diameters to examine sensitivity.

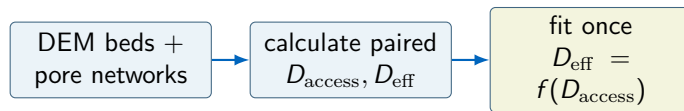
└ Reduction and diffusivity prediction

└ Development and later use are two different workflows

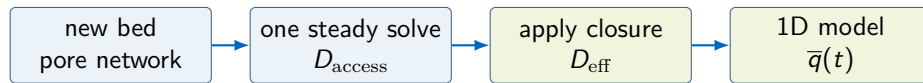
Development and later use are two different workflows



Develop closure



Predict for a new bed



For a new bed, the transient pore-network adsorption model is no longer required.

This summarizes the two stages we have now defined. During development, the transient PNM supplies reference uptake curves. Per-bed optimization supplies fitted diffusivities, and steady access calculations supply the descriptors. We learn a relationship between them. For another packing within the validated scope, we calculate its steady accessibility, apply the closure and solve the reduced one-dimensional system. That avoids repeating transient PNM adsorption and per-bed optimization. We now examine the evidence for each link in this workflow.

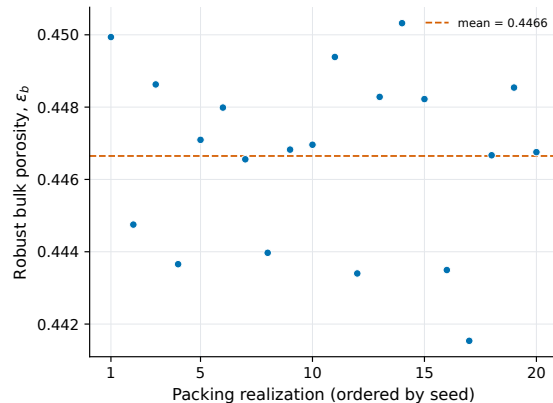
Section 4 of 5

Results and robustness

Test the physical hypothesis and the model reduction

Test the physical hypothesis and the model reduction.

The twenty beds have nearly equal bulk porosity



Bulk-porosity CV: 0.55%.

Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

CV: coefficient of variation

$$CV_{\varepsilon} = s_{\varepsilon} / \bar{\varepsilon}$$

s_{ε} : standard deviation

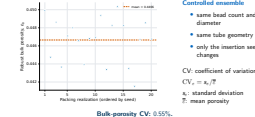
$\bar{\varepsilon}$: mean porosity

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Results and robustness

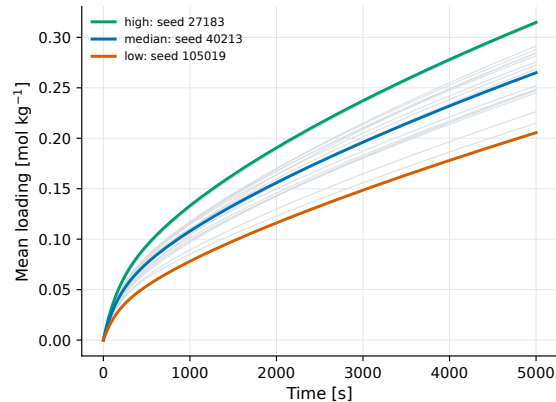
└ The twenty beds have nearly equal bulk porosity

The twenty beds have nearly equal bulk porosity



The mean bulk porosity is approximately 0.44665. Divide the sample standard deviation by this mean to obtain the coefficient of variation, 0.55 percent. The horizontal axis orders beds by their insertion seed. The small spread establishes the controlled comparison.

Similar porosity still produces different uptake curves



At 5000 s: 0.206–0.315 mol kg⁻¹, with an uptake CV of 11.0%.

Network reference

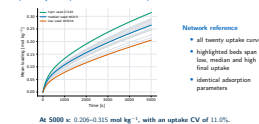
- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

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Results and robustness

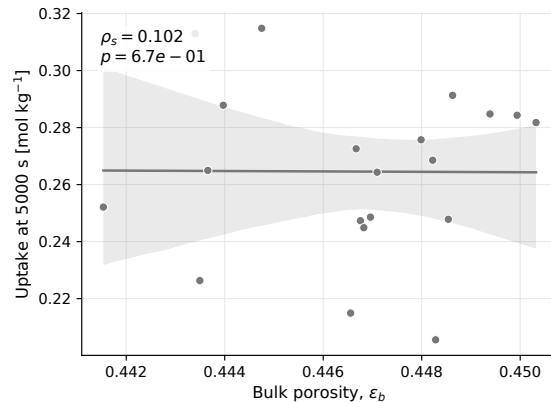
Similar porosity still produces different uptake curves

Similar porosity still produces different uptake curves



The pale curves show the full ensemble. Orange, blue and green identify low, median and high final uptake. These representative beds are selected by final uptake, not by accessibility. The uptake varies much more strongly than porosity.

Bulk porosity alone does not explain final uptake



Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

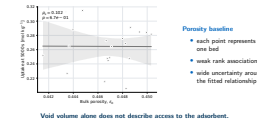
Void volume alone does not describe access to the adsorbent.

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Results and robustness

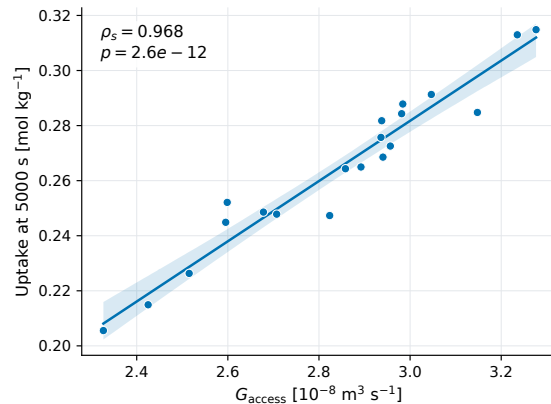
Bulk porosity alone does not explain final uptake

Bulk porosity alone does not explain final uptake



The plot compares bulk porosity with mean loading at 5000 seconds. Spearman correlation is approximately 0.102. The shaded band is pointwise bootstrap uncertainty in the fitted relationship, not a prediction interval for a new bed.

Accessibility explains differences in final uptake



Inlet-connected pathways explain variability that porosity misses.

Steady network descriptor

- G_{access} measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

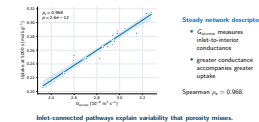
Spearman $\rho_s = 0.968$.

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Results and robustness

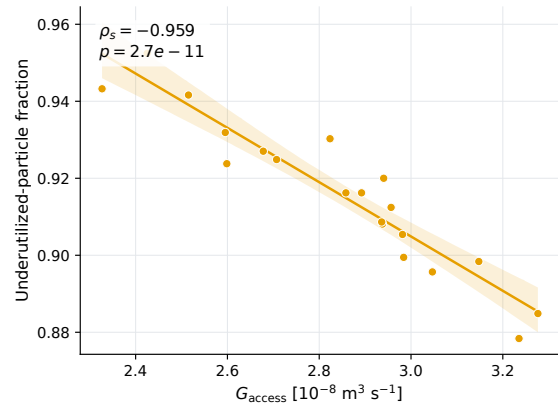
Accessibility explains differences in final uptake

Accessibility explains differences in final uptake



We calculate accessibility using a steady diffusion problem without adsorption. The Spearman coefficient is 0.968. More accessible beds generally have greater uptake. This is a within-ensemble association. The shaded band represents uncertainty in the fitted relationship.

Greater accessibility reduces particle underutilization



Higher G_{access} accompanies fewer underutilized particles.

Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

$I_k = 1$ when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

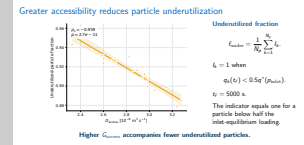
$t_f = 5000 \text{ s.}$

The indicator equals one for a particle below half the inlet-equilibrium loading.

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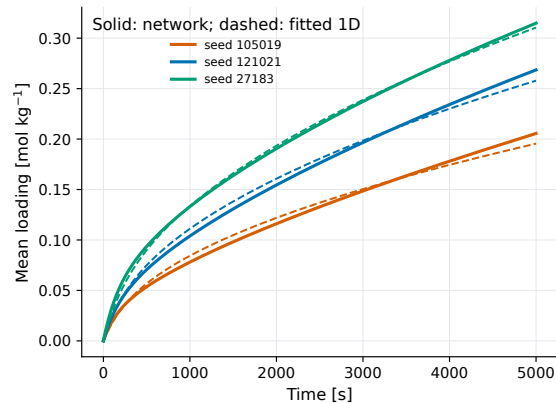
Results and robustness

Greater accessibility reduces particle underutilization



For each bead we compare its final loading with half the equilibrium loading at the inlet pressure. The indicator is one below this threshold and zero otherwise. Averaging over particles gives the plotted fraction. This is a threshold-based utilization measure, not the fraction of completely inaccessible particles.

One fitted D_{eff} reproduces each uptake curve



Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

Per-bed optimization

1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

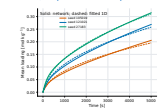
Solid: network.
Dashed: fitted 1D model.

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Results and robustness

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Per-bed optimization

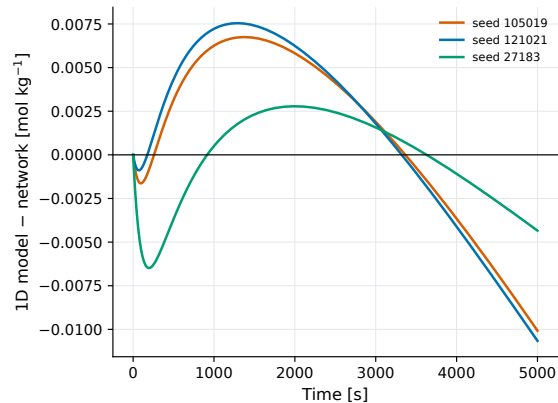
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Solid: network.
Dashed: fitted 1D model.

Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

For each bed, the transient pore-network calculation provides a reference uptake curve. We repeatedly solve the 1D model with trial diffusivities and minimize the time-weighted normalized root-mean-square error. The bounded optimizer searches in $\log_{10} D$. Its initial trial follows from the prescribed bounds, without random sampling. Previous objective values guide subsequent trials until the search meets its tolerance. Each trial solves both coupled equations from the same initial state. The optimized diffusivity is D_{eff} . This figure shows per-bed fits, not predictions for unseen beds. The mean error is 1.58 percent.

Residuals show the remaining reduction error



Time-weighted NRMSE across all beds: 0.84–2.46%.

Curve residual

$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

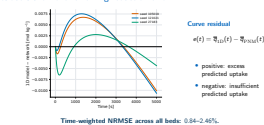
- positive: excess predicted uptake
- negative: insufficient predicted uptake

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Results and robustness

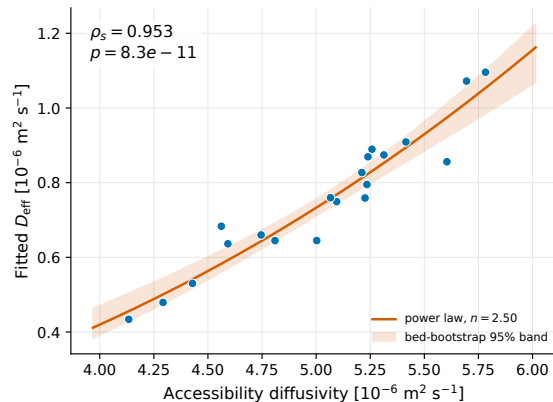
Residuals show the remaining reduction error

Residuals show the remaining reduction error



The same three representative beds appear here. The residual is the 1D mean loading minus the network mean loading. Its sign changes over time, showing that one coefficient cannot reproduce every detail. NRMSE summarizes the full residual curve using trapezoidal time weights and normalizes by final network loading.

A positive power law relates accessibility to diffusivity



The fitted exponent is $n \approx 2.50$ for this ensemble.

Selected closure

$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n$$

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}$$

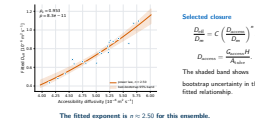
The shaded band shows
bootstrap uncertainty in the
fitted relationship.

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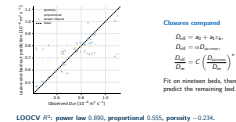
Results and robustness

A positive power law relates accessibility to diffusivity

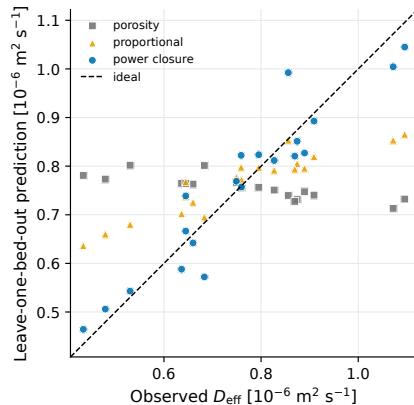
A positive power law relates accessibility to diffusivity



Each bed supplies a geometrically calculated accessibility diffusivity and a diffusivity fitted to its transient response. The power law relates them. This plot shows the fit to the complete ensemble. The band is pointwise bootstrap uncertainty in the relationship, not an individual-bed prediction interval. The exponent is specific to the studied conditions.



The closure predicts diffusivity for a withheld bed



Closures compared

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b,$$

$$D_{\text{eff}} = \alpha D_{\text{access}},$$

$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Fit on nineteen beds, then predict the remaining bed.

LOOCV R^2 : power law 0.890, proportional 0.555, porosity -0.234.

We leave out one entire bed, fit each closure to the other nineteen, and predict the omitted diffusivity. Repeating this gives twenty held-out predictions. Blue circles show the power law, orange triangles show proportionality, and grey squares show porosity. Model form and diagnostic depth were compared within this ensemble, so these scores remain internal validation.

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

Endpoint correlation summarizes one time. A transport closure supplies a complete uptake history.

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└ Results and robustness

└ Why not predict uptake directly from G_{access} ?

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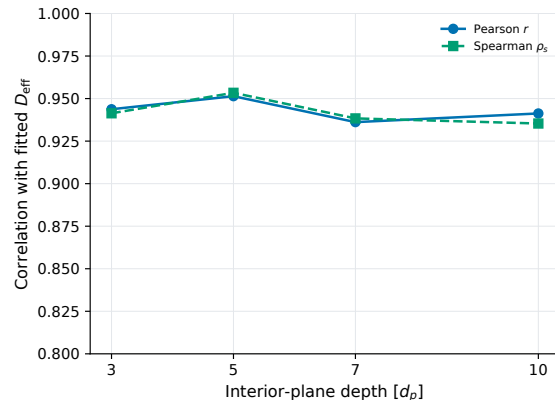
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Endpoint correlation summarizes one time. A transport closure supplies a complete uptake history.

The results show a strong relationship between accessibility and final loading. If only ranking at the selected final time is needed, that relationship may already be useful. It does not by itself give a full uptake history or validate transfer to other conditions. A closure for the reduced diffusivity instead feeds a dynamic model that produces the whole curve and axial fields. This is why our main objective is a transport closure rather than only an endpoint regression.

Accessibility remains correlated across diagnostic depths



Both correlations remain high across the tested depths.

Depth sensitivity

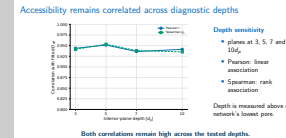
- planes at 3, 5, 7 and $10d_p$
- Pearson: linear association
- Spearman: rank association

Depth is measured above each network's lowest pore.

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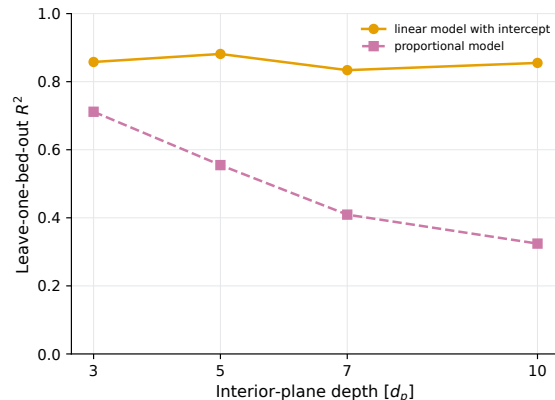
Results and robustness

Accessibility remains correlated across diagnostic depths



The original descriptor places the diagnostic plane above the lowest pore in each network. We tested three, five, seven and ten particle diameters using that convention. Both linear and rank correlations remain high. These depths are not measured exactly from the physical bed bottom.

Predictive performance is stable across diagnostic depths



Models in this comparison

- linear model with intercept

$$D_{\text{eff}} = aD_{\text{access}} + b$$

- proportional model

$$D_{\text{eff}} = \alpha D_{\text{access}}$$

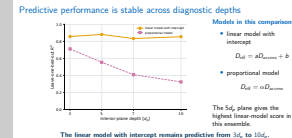
The $5d_p$ plane gives the highest linear-model score in this ensemble.

The linear model with intercept remains predictive from $3d_p$ to $10d_p$.

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Results and robustness

Predictive performance is stable across diagnostic depths



This plot compares affine and proportional closures at each depth. It does not plot the power-law score. Five particle diameters gives the best linear-model score among these tested depths. This sensitivity supports the association while independent beds are still needed to evaluate model and depth selection.

Section 5 of 5

Conclusions and next steps

What is established and what remains to validate

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What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

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└─ Conclusions and next steps

└─ What the present evidence establishes

What the present evidence establishes

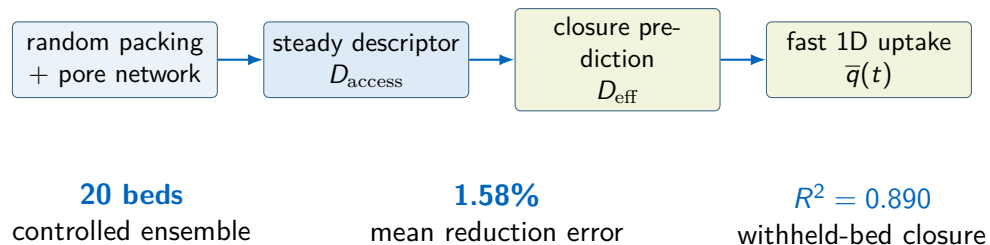
Supported	Not yet claimed
• porosity alone misses important packing variability • inlet-connected topology controls transient utilization • one D_{eff} reproduces each detailed curve accurately • D_{access} provides an interpretable predictive closure	• industrial breakthrough performance • effects of flow, heat or water competition • transfer to other bead sizes or D_{tube}/d_p • a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

The present evidence supports four conclusions. Bulk porosity misses important differences among random packings. Inlet-connected topology explains a large part of the resulting variation. One optimized D_{eff} allows the 1D model to reproduce each detailed uptake curve accurately. Finally, D_{access} provides a cheap and interpretable input for predicting that coefficient. The scope must remain clear. We have not modelled an industrial breakthrough column with imposed flow. We have not included heat effects, water or competitive adsorption. We also have not demonstrated that the fitted power law transfers to other bead sizes or tube-to-particle ratios. The result is a validated structure-to-model closure for this controlled, diffusion-driven adsorption problem, rather than a universal adsorption correlation.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



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Conclusions and next steps

Pore accessibility retains what bulk porosity loses

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Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



The main message is that equal void volume does not imply equal transport. How the void space connects to the inlet matters. Starting from a random packing, we extract the pore network and calculate the steady accessibility descriptor. The fitted closure converts this descriptor into D_{eff} , and the conservative 1D model uses that coefficient to predict the transient uptake.

Across twenty controlled beds, porosity varied by only 0.55 percent while uptake varied much more strongly. The optimized 1D model reproduced the pore-network curves with a mean NRMSE of 1.58 percent. The accessibility power law achieved a leave-one-bed-out R^2 of 0.890. The practical outcome is therefore a fast transient model that retains a clear and measurable link to the underlying packing structure.